

Potluck Planning

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Mode: Full Pipeline

1. Problem Formulation

Problem Type:	Find
Domain:	Integer Linear Programming
Named Problem:	Assignment Problem

Universe

- Assignment: $\{\text{Alice, Ben, Carmen, David, Elena, Frank, Grace, Hiro}\}$
- Category Counts: $\{\text{appetizer, salad, main, side, dessert}\}$

Variables

Symbol	Meaning	Type / Range
x_{ij}	assignment decision	$\{0, 1\}$
n	number of binary variables = 40	$\mathbb{Z}^+$

Structure

Assign 8 guests to 5 dish categories (appetizer, salad, main, side, dessert) for a potluck dinner. Each guest rates their effort/skill for each category on a 1-5 scale (lower = easier for them). Every category must be covered by at least 1 guest and at most 2 guests. Minimize total effort across all assignments.

Real-World Mapping

Real-World Concept	Mathematical Object
assignment	x : solution vector (assignment)

Constraints

(1) $\sum_j x_{ij} \leq 2$

capacity limit

(2) $\sum_j x_{ij} \geq 1$

minimum requirement

(3)

$$x_{ij} \in \{0, 1\} \quad \forall i, j$$

binary decision variables

Approach

Suggested approach: ILP (PuLP/CBC, Branch & Bound)

Complexity class:

Available tools: ILP

2. Solution

Answer:	8-element assignment: Alice -> dessert, Ben -> salad, Carmen -> main, D
Optimal:	Yes
Feasible:	Yes
Algorithm:	ILP (PuLP/CBC, Branch & Bound)
Time:	0.0059s
Certificate:	Branch-and-bound solved to optimality.

Solution Details

Assignment:

Alice -> dessert
Ben -> salad
Carmen -> main
David -> side
Elena -> appetizer
Frank -> dessert
Grace -> salad
Hiro -> appetizer

Method: Integer Linear Programming (ILP) via PuLP/CBC.

- 40 binary variables (8 guests x 5 categories)
- Constraints: one category per guest, min/max coverage per category
- Objective: minimize sum of effort scores across all guest-category assignments

Verification

✓ Feasibility	All constraints satisfied
✓ Optimality	ILP (PuLP/CBC, Branch & Bound)

3. Interpretation

The Question

Assign 8 guests to 5 dish categories (appetizer, salad, main, side, dessert) for a potluck dinner.

The Answer

A optimal solution was found.

What This Means

- Optimal assignment of guests to dish categories minimizing total effort
- All 5 categories covered (some by 2 guests since $8 > 5$)
- Independent constraint verification (all PASS)
- Comparison against theoretical bounds and random assignment

Integer Linear Programming (ILP) via PuLP/CBC.

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Recommendations

1. The solution is provably optimal; no further improvement is possible within this model.

Limitations

- Model assumes deterministic parameters; real-world variability not captured.

• Solution